Shaojie Hu

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Research Profile

My research focuses on the thermodynamics and phase transitions of water in porous media. I am interested in understanding how these pore-scale processes shape Earth and engineering systems at the large scale. My work has been published in journals including ***Physical Review Letters***, ***Water Resources Research***, and ***Géotechnique***. I also serve as an ***Associate Editor*** of ***Vadose Zone Journal***.

Current research interests include:

- **Phase Transitions in Porous Media:** Vapor sorption and capillary condensation; pore-water freezing and melting; pore-salt deliquescence and efflorescence.
- **Interfacial Thermodynamics and Transport:** Soil-water interactions; thin-film and confined water transport; stable-isotope fractionation of pore water.
- **Soil Physics and Poromechanics:** Soil water potential; soil water retention and hydraulic conductivity; effective stress; hydro-mechanical behavior of unsaturated soils.

Education

Ph.D., Civil Engineering Sept. 2020 – Jul. 2026
Hunan University, Changsha, China
Supervisor: Prof. Chao Zhang
Thesis: Unsaturated Mechanical Behaviors of Salt-Affected Soils

B.S., Civil Engineering Sept. 2016 – Jun. 2020
Hunan University, Changsha, China
Supervisor: Prof. Chao Zhang

Publications

Journal Articles (* Corresponding author)

[1] **Hu, S.**, Zhao, N., Zhang, C.*, Li, F., Chen, R., & Or, D.* (2025). Water nanofilms facilitate ice crystal growth across droplets. *Physical Review Letters*, *134*(6), 064001.

[2] Zhang, C., Zhao, N., **Hu, S.***, & Lin, X. (2025). Salts retard ice crystal growth in supercooled droplets during recalescence. *Physical Review Fluids*, *10*(6), 063605.

[3] **Hu, S.**, Zhang, C.*, Li, L., Or, D., Lu, N., & Chen, R. (2026). Fractionation due to soil–water interactions affects global isotopic evapotranspiration partitioning. *Advances in Water Resources*, 105406.

[4] **Hu, S.**, Zhang, C.*, & Lu, N. (2023). Quantifying coupling effects between soil matric potential and osmotic potential. *Water Resources Research*, *59*(2), e2022WR033779.

[5] Zhang, C., Gou, L., **Hu, S.***, & Lu, N. (2022). A thermodynamic formulation of water potential in soil. *Water Resources Research*, *58*(9).

[6] Zhang, C., **Hu, S.**, Qiu, Z., & Lu, N.* (2022). A poroelasticity theory for soil incorporating adsorption and capillarity. *Géotechnique*, 1–18.

[7] Zhang, C., Lin, X., **Hu, S.***, & Lu, N. (2025). Predicting the soil freezing curve via a modified Clapeyron equation. *ASCE Journal of Geotechnical and Geoenvironmental Engineering*, *152*(2), 04025195.

[8] Zhang, C.*, **Hu, S.**, & Lu, N. (2022). Unified elastic modulus characteristic curve equation for variably saturated soils. *ASCE Journal of Geotechnical and Geoenvironmental Engineering*, *148*(1), 04021171.

[9] Gou, L., Lu, N., **Hu, S.**, Calderon, A. R. A., & Zhang, C.* (2025). Suction stress of soil slurry. *ASCE Journal of Geotechnical and Geoenvironmental Engineering*, *151*(2), 04024164.

[10] Gou, L., Zhang, C.*, Lu, N., & **Hu, S.** (2023). A soil hydraulic conductivity equation incorporating adsorption and capillarity. *ASCE Journal of Geotechnical and Geoenvironmental Engineering*, *149*(8), 04023056.

[11] **Hu, S.**, Zhang, C.*, Li, L., & Lu, N. (2026). Determining soil water structure, density, and viscosity via infrared spectroscopy. *ASCE Journal of Engineering Mechanics*, *152*(4), 04026002.

[12] **Hu, S.**, Zhang, C.*, Dong, Y., Gou, L., & Chen, R. (2025). Water vapor sorption isotherms of salt-affected soils. *Canadian Geotechnical Journal*, cgj-2024-0426.

[13] **Hu, S.**, & Zhang, C.* (2023). A sorption isotherm model for soil incorporating external and internal surface adsorption, and capillarity. *Canadian Geotechnical Journal*, cgj-2022-0386.

[14] Lin, X., Zhang, C.*, **Hu, S.**, & Chen, R. (2024). Heterogeneous ice nucleation of salt solution in porous media. *The Journal of Chemical Physics*, *160*(9), 094501.

[15] Zhang, C., Li, L., **Hu, S.***, Gou, L., & Chen, R. (2024). Physical origin of adsorption heat and its significance in the isotherm equation. *International Journal of Heat and Mass Transfer*, *220*, 124914.

[16] Zhao, N., **Hu, S.**, Zhang, C.*, Li, F., & Chen, R. (2023). Physical origins of freezing and melting temperature depressions of water in millimeter-sized pores. *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, *674*, 131851.

[17] Gou, L., Zhang, C.*, **Hu, S.**, Chen, R., & Dong, Y. (2023). Semi-analytical solutions for soil consolidation induced by drying. *Acta Geotechnica*, *18*(2), 739–755.

Manuscripts under review and in preparation

[18] **Hu, S.**, Zhang, C.*, Gou, L., & Chang, J. Activation Energy Model for Film Flow Incorporating Adsorption. *Under review.*

[19] Thermodynamics of Equilibrium Isotope Fractionation of Water in Porous Media. *In preparation.*

[20] Global Water Ice Content Mapping in Martian Regolith with Refined Infrared Inversion. *In preparation.*

Presentations

Gordon Research Conference: Flow and Transport in Permeable Media
Les Diablerets, Switzerland — Jul. 2026
Poster: Fractionation due to soil–water interactions affects global isotopic evapotranspiration partitioning

AGU25, New Orleans, LA, United States — Dec. 2025
Poster: Fractionation Due to Soil–Water Interactions Is an Important Omission in Global Isotopic Evapotranspiration Partitioning

Ice at the Micro-scale 2025, Ascona, Switzerland — July 2025
Poster: Salts retard ice crystal growth in supercooled droplets during recalescence

AGU24, Washington, D.C., United States — Dec. 2024
Poster: Phase equilibrium and transitions in salt-affected soils

InterPore2024, Qingdao, China — May 2024
Oral: Adsorption-induced Effective Stress in Porous Media

The 4th International Soil Modeling Consortium Conference, Tianjin, China — May 2024
Poster: Coupling Between Soil Matric Potential and Osmotic Potential

InterPore2023, Edinburgh, Scotland — May 2023
Poster: A Poroelasticity Theory for Soil Incorporating Adsorption and Capillarity

The 9th Young Experts Forum on Geotechnical Engineering, Changsha, China — Jun. 2021
Oral: Unified Elastic Modulus Function for Variably Saturated Soils

Academic Service

Editorial Service

Associate Editor, *Vadose Zone Journal* — 2026 – present

Social Media Editor, *Vadose Zone Journal* — 2025 – 2026

Journal Reviewer

Communications Earth & Environment; Water Resources Research; Vadose Zone Journal; Geoscientific Model Development; Bulletin of Engineering Geology and the Environment; Construction and Building Materials; Geotechnical Research.

Awards & Honors

First Prize of AGU Student Award — 2024
Unsaturated Zone Technical Committee, American Geophysical Union

National Scholarship for Graduate Students — 2024
Ministry of Education, China

Engineering Innovation Design Scholarship — 2022
Hunan University

Outstanding Innovative Undergraduate Thesis — 2020
Hunan University

Technical Skills

Computational Methods and Programming: MATLAB, Python, QGIS, and molecular dynamics simulation.

Experimental Techniques: Water vapor sorption, infrared spectroscopy, soil strength and deformation testing; experience with synchrotron X-ray scattering (WAXS/SAXS) and neutron scattering experiments.